# VO XUAN NHI TRAN

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## ABOUT ME

Third-year Software Engineering student at the University of Sydney with hands-on experience building and deploying full-stack applications and AI-powered tools, including a live job-ad screener that drew about 35 visitors and 40+ analyses in its first two weeks. Skilled in Python, TypeScript, React, FastAPI, PostgreSQL and AWS, with practical experience in automated testing with a real client and cloud deployment. Interested in building reliable software that solves real-world problems.

## EDUCATION

**University of Sydney** July 2024 – Present | Sydney, Australia

**Bachelor of Engineering Honours (Software Engineering) - Dalyell Scholar - Vice Chancellor's International Scholarship**

- **WAM:** 77.75/100 (Distinction Average) | **Expected Graduation:** 2028
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Agile Software Development, Security Engineering, Software Construction and Design, Data and Information Management

## PROJECTS

**JobCompass (Futura Remix Hackathon 2026)** GitHub | Live Demo

*Technologies: Next.js, TypeScript, Gemini API, Upstash Redis, PostHog, Tailwind CSS, Vitest*

- Built JobCompass, a visa-eligibility screener for international students: paste a job ad and get an explainable APPLY / TAILOR / SKIP verdict backed by the exact sentence from the ad.
- Paired Gemini evidence extraction with a deterministic rules engine; any claim without a supporting quote is dropped, so the model never decides the verdict itself.
- Evaluated on a fresh set of 20 job ads I labelled before running: 15/20 correct (SKIP 7/7, APPLY 7/9, TAILOR 1/4), with no blocked job marked APPLY. Traced the 5 misses to 4 failure patterns, mostly in extraction. Separately kept 35 dev and 15 hold-out ads as regression tests (35/35 and 14/15).
- Shipped a live app (Next.js, Vercel, Upstash Redis) with PostHog analytics, a 20s model timeout with a graceful fallback, rate limits and caching; about 35 people opened it in its first two weeks and it has run 40+ job-ad analyses.

**RAG Document Assistant** GitHub | Live Demo

*Technologies: Python, FastAPI, React, PostgreSQL/pgvector, Docker, AWS*

- Built a full-stack RAG app (React, FastAPI, PostgreSQL/pgvector, Gemini API) where users upload PDFs and get answers grounded in their documents; deployed on AWS EC2 with Nginx and CI/CD, debugging CORS and HTTPS mixed-content issues along the way.
- Wrote 13 pytest tests with mocked LLM calls and an isolated test database, including cross-session isolation tests that verify one user's documents are never retrievable by another.

**RenoPilot – Roof Replacement App | Capstone Project with Real Industry Client in team of 8**

*Technologies: Next.js, React, TypeScript, Payload CMS, PostgreSQL*

- Built the four-option landing page by extending the shared AppLanding component without breaking the other RenoPilot apps, then added persistent light/dark themes across the quote workflow (merged, client-accepted).
- Regression-tested the Property → Map → Materials → Quote flow and Undo feature (148 tests passing), reporting bugs to feature owners until fixed.
- Rotated through Tester Lead, Programmer Lead and Customer Liaison, presenting progress to the client.

**YouTube Video Dubber (Chinese to Vietnamese)** GitHub

*Technologies: Python, faster-whisper, Gemini API, ffmpeg*

- Automated a Chinese-to-Vietnamese dubbing pipeline (faster-whisper transcribe, Gemini translate, TTS, ffmpeg merge).
- Fixed a 35% audio overrun with a word-budget translation step based on measured TTS timings and silence trimming, reaching 0.00s drift across 13 blocks on a 72-second clip.
- Added caching and retry/backoff to work around Gemini free-tier quota limits.

## TECHNICAL SKILLS

- **Languages:** Python, Java, TypeScript/JavaScript, SQL, HTML/CSS
- **Frameworks:** React, Next.js, FastAPI, Django, Django REST Framework
- **Databases:** PostgreSQL, pgvector, SQLite
- **Cloud & DevOps:** Docker, AWS EC2, Nginx, Vercel, Render, Linux
- **Tools:** Git/GitHub, pytest, JUnit, Gemini API